

db_config.php

PHP

```
$conn = mysqli_connect("localhost", "root", "", "student_db");
if (!$conn) die("Connection failed: " . mysqli_connect_error());
```

Insert/Display Logic (index.php)

PHP

```
include 'db_config.php';

// Create (Insert)
if(isset($_POST['add'])){
    $name = $_POST['name'];
    $sql = "INSERT INTO students (name) VALUES ('$name')";
    mysqli_query($conn, $sql);
}

// Read (Select)
$result = mysqli_query($conn, "SELECT * FROM students");
while($row = mysqli_fetch_assoc($result)){
    echo $row['name'] . " <a href='delete.php?id=".$row['id']."'>Delete</a><br>";
}
```

2. Electricity Bill Calculation (Question 6)

logic.php

PHP

```
function calculate_bill($units) {
    $bill = 0;
    if($units <= 50) $bill = $units * 3.50;
    else if($units <= 150) $bill = (50 * 3.50) + (($units - 50) * 4.00);
    else if($units <= 250) $bill = (50 * 3.50) + (100 * 4.00) + (($units - 150) * 5.20);
    else $bill = (50 * 3.50) + (100 * 4.00) + (100 * 5.20) + (($units - 250) * 6.50);
    return $bill;
}
```

3. React State & Props (Questions 7, 8, 16, 28, 29)

Use this structure for your React-based UI tasks.

App.js (Theme Toggle/Clock Example)

JavaScript

```
import React, { useState, useEffect } from 'react';

function App() {
    const [time, setTime] = useState(new Date().toLocaleTimeString());
    const [theme, setTheme] = useState('light');
```

```

useEffect(() => {
  const timer = setInterval(() => setTime(new Date().toLocaleTimeString(), 1000);
  return () => clearInterval(timer);
}, []);

const style = {
  background: theme === 'light' ? '#fff' : '#333',
  color: theme === 'light' ? '#000' : '#fff',
  height: '100vh'
};

return (
  <div style={style}>
    <h1>{time}</h1>
    <button onClick={() => setTheme(theme === 'light' ? 'dark' : 'light')}>
      Toggle Mode
    </button>
  </div>
);
}
export default App;

```

4. Node.js & Express REST API (Questions 10, 27, 34, 35)

server.js

JavaScript

```

const express = require('express');
const app = express();
app.use(express.json());

let students = []; // Mock DB

app.post('/register', (req, res) => {
  const student = req.body;
  students.push(student);
  res.status(201).json({ message: "Student Registered" });
});

app.get('/students', (req, res) => {
  res.json(students);
});

app.listen(3000, () => console.log("Server running on 3000"));

```

5. Spring Boot Entity & Repository (Questions 9, 14, 26)

Order.java

Java

```

@Entity
public class Order {
  @Id @GeneratedValue(strategy = GenerationType.IDENTITY)
  private Long id;
  private String productName;
  // Getters and Setters

```

```
}
```

OrderRepository.java

Java

```
public interface OrderRepository extends JpaRepository<Order, Long> { }
```

6. JavaScript Array Operations (Question 3)

JavaScript

```
let arr = [40, 10, 100, 1];  
// Reverse  
arr.reverse();  
// Sort (Numeric)  
arr.sort((a, b) => a - b);  
// Search  
let index = arr.indexOf(100);
```

Tips for your Paper Work:

- **HTML:** Always include the `<!DOCTYPE html>` tag and link Bootstrap via CDN for responsive tasks.
- **PHP:** Remember to start with `<?php` and use `session_start()` at the very top for any session-related questions (Q4, Q11, Q13).
- **Bootstrap Grid:** Use `<div class="container"><div class="row"><div class="col-md-6">...</div></div></div>` for multi-column layouts.
- **MySQL:** For question 5 and 10, write the SQL queries clearly: `CREATE DATABASE student_db;` and `CREATE TABLE students (id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(50), email VARCHAR(50));`.

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To make your paper-writing faster, here is the **Standard HTML5 Boilerplate** including Bootstrap and jQuery. Most of your questions (1, 2, 4, 6, 12, 13, 15, 21) require this structure.

The "Base" HTML Template

Write this first for any frontend or PHP task. It includes the necessary CDNs for Bootstrap (Styling) and jQuery (Scripting).

HTML

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <title>Practical Exam</title>  
  <!-- Bootstrap CSS -->  
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">  
<style>  
  /* Common CSS for layouts (Question 1, 6) */  
  .header-section { padding: 50px 0; text-align: center; }  
  .custom-card { margin-top: 20px; padding: 20px; border-radius: 10px; }
```

```

</style>
</head>
<body class="bg-light">

  <div class="container mt-5">
    <!-- Content Goes Here -->
  </div>

  <!-- jQuery and Bootstrap JS -->
  <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

Key Snippets for Specific Tasks

1. Responsive Form (Question 4, 5, 10, 13)

Use this inside the container div.

HTML

```

<div class="card p-4 shadow">
  <form action="process.php" method="POST">
    <div class="mb-3">
      <label>Name:</label>
      <input type="text" name="name" class="form-control" required>
    </div>
    <div class="mb-3">
      <label>Email:</label>
      <input type="email" name="email" class="form-control" required>
    </div>
    <button type="submit" name="submit" class="btn btn-primary w-100">Submit</button>
  </form>
</div>

```

2. Bootstrap Grid (Question 1 - Multi-column)

HTML

```

<div class="row">
  <div class="col-md-4">Column 1 Content</div>
  <div class="col-md-4">Column 2 Content</div>
  <div class="col-md-4">Column 3 Content</div>
</div>

```

3. Bootstrap Tabs/Pills (Question 2)

HTML

```

<ul class="nav nav-pills mb-3" id="pills-tab" role="tablist">
  <li class="nav-item">
    <button class="nav-link active" data-bs-toggle="pill" data-bs-target="#sdp">SDP</button>
  </li>
  <li class="nav-item">
    <button class="nav-link" data-bs-toggle="pill" data-bs-target="#edi">EDI</button>
  </li>
</ul>
<div class="tab-content">
  <div class="tab-pane fade show active" id="sdp">SDP Content here...</div>
  <div class="tab-pane fade" id="edi">EDI Content here...</div>
</div>

```

This Bootstrap structure creates a **dynamic interface** where clicking a button (pill) swaps the visible content without refreshing the page.

1. The Navigation (The Buttons)

The `<ul class="nav nav-pills">` acts as the container for your clickable triggers.

- * `data-bs-toggle="pill"`: This is a JavaScript instruction for Bootstrap. It tells the browser, "When this button is clicked, toggle the visibility of a tab pane."
- * `data-bs-target="#sdp"`: This links the button to the specific content area. The value must match the `id` of the content div you want to show.
- * `active` Class: This highlights the button (usually with a blue background) to show which tab is currently selected.

2. The Content (The Panes)

The `<div class="tab-content">` holds all the possible information sections.

- * `.tab-pane`: Each child div inside the container represents one "page" of content.
- * `id="sdp"` / `id="edi"`: These are unique identifiers that the buttons look for.
- * `.fade`: This adds a smooth transition effect when switching between tabs.
- * `.show .active`: On the initial page load, only one pane can be visible. These classes tell Bootstrap which content to display by default.

How it works in the browser:

1. **Initial State**: The "SDP" button is blue, and the "SDP Content here..." text is visible. The "EDI" content is hidden via CSS (`display: none`).
2. **The Interaction**: When you click the "EDI" button, jQuery/Bootstrap JS listens for that click.
3. **The Switch**:
 - * It removes the `.active` class from the SDP button and pane.
 - * It adds the `.active` and `.show` classes to the EDI button and pane.
 - * The UI updates instantly to show the new content.

Why use this for your VIT Practical?

It is the most efficient way to organize different categories (like SDP, EDI, DT) on a single page, keeping the layout clean and "responsive" (as it stacks well on mobile).

4. Simple Navigation Bar (Question 14, 15)

HTML

```
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container">
    <a class="navbar-brand" href="#">VIT Portal</a>
    <div class="navbar-nav">
      <a class="nav-link" href="home.php">Home</a>
      <a class="nav-link" href="login.php">Login</a>
    </div>
  </div>
</nav>
```

Pro-Tips for your Paper:

- **Attributes:** Don't forget `name=""` in input tags for PHP tasks (that's how `$_POST['name']` works).
- **Methods:** For Question 4, use `method="GET"` or `method="POST"` depending on the task requirement.
- **Validation:** For Email, just use `type="email"` in HTML5; it validates automatically.